

Yizhi Chen

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EDUCATION

Georgia Institute of Technology <i>M.S : Computer Science GPA 4.0 — Networking, Cybersecurity, Advanced OS, ML</i>	Atlanta, GA, U.S. 01/2025 - flexible end date
Trine University <i>M.S : Information Science GPA 3.9 — Advanced Database, Data Science, Data Mining and Visualization</i>	Allen Park, MI, U.S. 06/2022 - 06/2023
China University of Petroleum <i>B.S : Software Engineering — Horn Scholarship Award — Computer architecture, Compiler, AI, Cloud</i>	09/2016 - 06/2020

TECHNICAL SKILLS

Backend: C/C++, Python (FastAPI, Flask, Django), Go (Gin, Go-zero), Node.js (Koa2)
Databases / Middleware: PostgreSQL, MySQL, Redis, MongoDB, Kafka, ElasticSearch, Databricks, Redis
DevOps & Cloud Native: AWS, Azure, Docker, K8s, Terraform, CI/CD(Jenkins, Argo)
AI / ML Ops: LLM Integration (OpenAI API), LangChain, Text-to-SQL, Streamlit
Frontend: TypeScript, React, Next.js, React Native, JavaScript (ES6+), Jotai, Tailwind CSS, D3.js, Webpack, Vite

EXPERIENCE

Software Engineer Intern <i>Nvidia @Cloud Cost Management Team</i>	05/2025 - Current Santa Clara, CA, U.S.
- Built the Cloud Cost Management platform that automated financial data aggregation and multi-cloud analysis (AWS, GCP, Azure, OCI) across all company organizations, reducing the finance reporting cycle from weeks to days. - Contributed to the scalable backend service with Python (FastAPI) and PostgreSQL to handle terabytes of financial data from 4 major CSPs. Automated a real-time, low-latency analytics pipeline aggregating company-wide cloud cost data, enabling finance team to optimize costs and save over \$10M+ annually. - Containerized and deployed the platform on K8s. Automated a large-scale data synchronization pipeline from Databricks with Delta Live Tables and Warehouse, ensuring real-time, cross-microservice data consistency and reliability. - Designed and implemented a robust RBAC system leveraging JWT and Azure AD for secure authentication and authorization. Deployed the IAM audit logging and alerting pipeline, improving security visibility across services. - Designed and implemented a Text-to-SQL AI service with LangChain and OpenAI APIs, successfully automating over complex user queries and reducing manual support tickets. - Build React/TypeScript frontend from scratch, leveraging React-Query and Jotai for data fetching and management.	
Software Engineer Intern <i>EZdealer</i>	02/2023 - 05/2023 Remote, U.S.
- Developed the responsive frontend with Vue, Webpack and Gulp, supported PC and mobile version of web app. Improve user engagement and cross-platform performance. - Enhanced front-end performance through Tree-shaking and Module Federation. Strengthened cache-update mechanisms by appending hash suffixes to output files, improving load speed and deployment reliability. - Developed and maintained the internal DMS(Dealer Manage System), resolving 30+ legacy bugs and enhancing system stability to reduced maintenance and support effort.	
Software Engineer <i>Xiaoice.AI (AI Company split from Microsoft) @Could Platform Team</i>	06/2021 - 07/2022 BeiJing, China
- Developed and maintained backend services for a suite of AI-powered applications with Python (Flask) and MongoDB, supporting over 10K+ DAU and achieving a Top 100 ranking in the Apple App Store. - Designed and built the cloud-native business intelligence monitoring platform on Azure that ingested and processed large-scale business data stream with Elasticsearch, Azure SQL Database, Blob Storage, AI&ML service. - Implement application modules to cloud-native backend services through Azure Functions, leveraging API triggers, queue bindings, and blob storage integration to handle asynchronous workloads. - Published Vue and React components libraries with Webpack, accompanied by comprehensive documentation written in Vue-press and Storybook to enhance the readability of the onboarding and deployment process for the AI platform. Published the library in Azure Artifacts, supporting easy creation of AI projects, with over 1k+ users.	
Software Engineer <i>SENSORO. @Machine Vision Team</i>	06/2020 - 06/2021 BeiJing, China
Developed a scalable IVMS(Intelligent Video Management System) with Python Tornado, MySQL, and Elasticsearch, supporting multi-country i18n and delivering real-time search and analytics across distributed video datasets	

- Developed event-driven data labeling backend in **Golang** with **gRPC APIs**, implementing **computer vision labeling pipelines** for the Visualization Platform. Enabled automated annotation, real-time tracking, and quality validation, boosting QA annotation efficiency by **24%**
- Implemented **API and integration test** with **Koa2** and **Jest in Node.js**, resulting in a test coverage increase to over **95%**, guaranteeing thorough and reliable testing of the service.
- Deployed and maintained large-scale SENSORO sensor networks on **Alibaba Cloud**, enabling real-time aggregation and processing of **IoT telemetry MQTT (Message Queuing Telemetry Transport)**. Implemented monitoring system for device failures, improving operational reliability and reducing downtime across multi-region deployments
- Managed and automated service deployments with **Docker**, **K8s**, **Jenkins**, implemented robust CI/CD pipelines for **multi-region, cloud-native releases**. Improved deployment reliability, reduced release time.

Software Engineer Intern

12/2019 - 06/2020

SENSORO @Cloud Platform Team

BeiJing, China

- **Built a secure video delivery system** with **video.js** and **FFmpeg**, enabling encrypted streaming and preventing unauthorized distribution. Deployed on **Alibaba Cloud** with **CDN distribution** and **object storage** integration.
- **Improved SEO**, boosting search rankings by **20%**. Added **Google Analytics** tracking and **reCAPTCHA** verification to strengthen security. Built a **serverless form** with **Formspree**, removing backend dependencies. **Tech Stack: Gulp, jQuery, Google Analytics, Formspree, Google reCAPTCHA.**

PROJECTS

Artify-AI

AI-powered drawing platform featuring Text-to-Image generation, background removal, secure OAuth login, and Stripe-integrated online checkout. Tech stack: Next.js, TypeScript, Shadcn/ui, React Query, Zustand, Zod, Fabric.js, Hono, OAuth. - [Source Code](#)

Finance Server in Go

Secure finance service in Go, implementing gRPC APIs, Paseto authentication, PostgreSQL with sqlc, Redis-powered background jobs, and automated migrations. - [Source Code](#)

Cloud Efficiency Toolkit

Backend toolkit in Python with Terraform and shell automation to analyze cloud resource usage, optimize costs, and streamline infrastructure management. Cloud Support: AWS, Azure - [Source Code](#)

Distributed File System with gRPC

Distributed file system with advanced proxy-server for large-file caching, auto synchronization, and reliable multi-threading client-server file handling. Tech: Built with C/C++, gRPC. [Source Code](#)

PUBLICATIONS

- [Research on Design and Application of Mobile Edge Computing Model Based on SDN.](#)
doi: 10.1109/ICCCN49398.2020.9209601.
- [Improvements and Challenges in StarCraft II Macro-Management A Study on the MSC Dataset.](#)
doi: 10.53469/jtpes.2023.03(12).05.
- [Computer Vision and Deep Learning Transforming Image Recognition and Beyond.](#)
doi: 10.62051/ijcsit.v2n1.06